SIMATSENGINEERING(SAVEETHASCHOOLOFENGINEERING)

DepartmentofComputerScienceandEngineering

ASSESSMENTTOOL3–SHORTANSWERTEST

CourseCode:	CSA0305	**CourseTitle:**	DataStructures
COAssessed:	CO5–Manipulation(BL3,BL4)	**Assessment:**	AssessmentTool3–ShortAnswer Test
Weightage:	30%ofCIA	**TotalMarks:**	100
CO2Statement:	*ConstructMSTusing shortest pathalgorithms andtraverse the graphto model and analysereal-world problem.*		
StudentName:	M.VAISHNAV	**Reg. No.:**	192524251
Section/Batch:	2025	**Date:**	04/09/2026

CodeofConduct

I __M.VAISHNAV____(Name/RegNo) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature ofthe Student: M.VAISHNAV

Instructions

- Thistestcontains10shortanswertestquestions.Total:100Marks.
- WhenastudentanswerusinganAnalyticalproblem-solving,inevaluationtheanswershouldbewith followingfiverequirements:Understandingofproblem,selectappropriatedatastructure,Designthe solution, analyze, Clarity of representation.
- Nonegativemarking.Unansweredquestionscarryzeromarks.
- NoDTPprintoutsorelectronicdevicespermitted.Theyshouldprovidewrittenmaterials.

Section/Topic	Focus	QNos	Marks
MinimumSpanningTree	Kruskal'sAlgorithm	1	20
Graph Sorting	TopologicalSort	2	20
ShortestPathAlgorithm	MST&Dijkstra'sAlgorithm	3	20
ShortestPathAlgorithm	Dijkstra'sAlgorithm	4	20
MinimumSpanningTree	Prim'sorKruskal'sAlgorithm	5	20
	GRANDTOTAL:	100Marks	

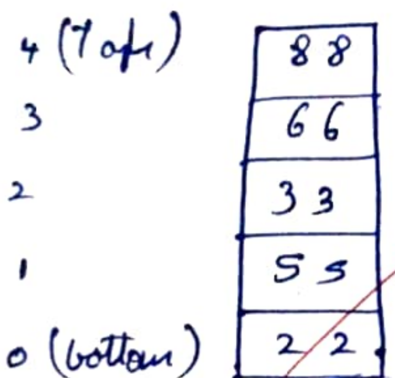
Criteria	Weightage	Level4-Exemplary	Level3-Proficient	Level 2 - Developing	Level 1 - Beginning
ConceptualUnderstanding	30	Demonstratescomplete andaccurateunderstand ingofconcepts	Goodunderstanding with minor gaps	Basicunderstanding	Limitedundersta nding
Accuracyof Answer	25	Answersarecompletelyco rrect and relevant	Mostlycorrectwithm inorerrors	Partiallycorrect	Mostlyincorrect
Problem-Solving&Reasoning	20	Provideslogicalre asoningandappro priatesteps	Goodreasoningwithmin orgaps	Basicreasoning	Lackslogical reasoning
Clarity&Relevance	15	Answersareprecise,clea r, concise, anddirectly address thequestion	Mostlyclearandrelev ant	Somewhatunclearorinc omplete	Irrelevantorunclearanswers
TimeManagement&Completion	10	Completesallquestionsa ccurately within theallottedtime	Completesmostques tionsontime	Requiresadditional time	Unabletocompletethe test

FacultyIn-charge	CourseCoordinator	ProgramDirector
Signature:	Signature:	Signature:
Date:	Date:	Date:

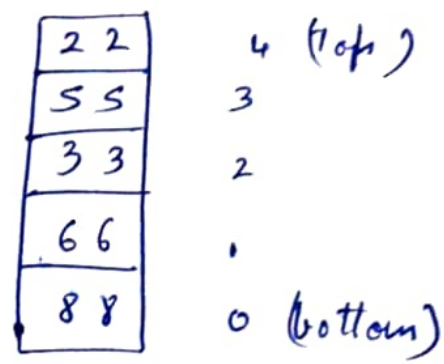
1) Perform the following operations using stack. Assume the size of the stack is 5 and have a value of 22, 55, 33, 66, 88 in the stack from 0 position to size-1. Construct ALL tree for the following elements.

Ans: intial stack (size = 5)

Initial stack.



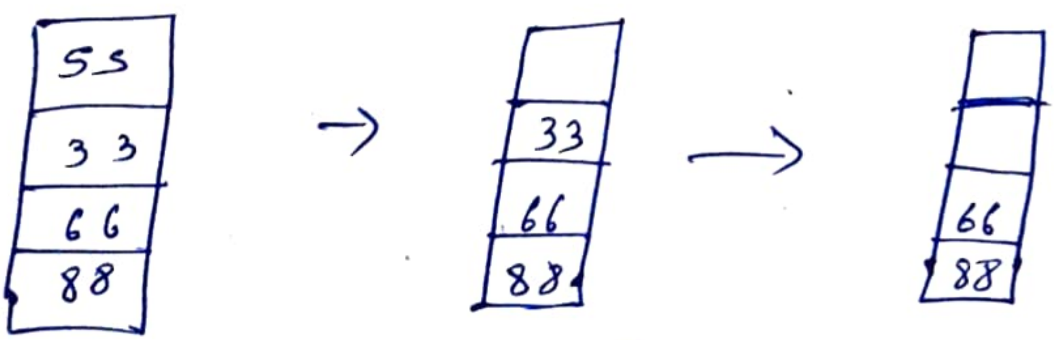
1) Insert the Elements



2) Pop() → 22

3) POP() → 55

4) POP() → 33



⇒ Remaining stack. bottom top : 88, 66

2) Linear Search Vs Binary Search

Ans: Elements 10, 20, 30, 40, 50, 60, 70, 80

Key = 70

⇒ Linear Search

10	20	30	40	50	60	70	80

Comparisions

(10 → 20 → 30 → 40 → 50 → 60 → 70)

Time Complexity : O(n)

⇒ Binary Search

mid ↓ (at 40)

10	20	30	40	50	60	70	80
0	1	2	3	4	5	6	7

40 < 70 . Go right

mid ↓

50	60	50	30	80
4	5	6	6	7

60 < 70 Go right

mid ↓

70	80
6	7

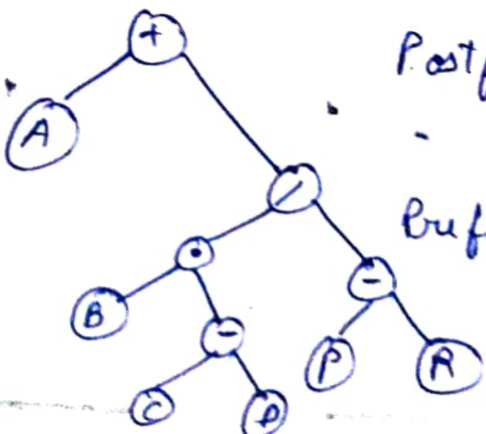
70 = 70 Found.

Comparisons: 3

Time Complexity : O(log n)

3) Prefix And POSTFIX

Ans: Expression A + B * (C - D) /(P - R)



Postfix: AB CD - * PR - / +

Prefix: + A / * B - C D - P R

Merge Sort

Input : 40, 20, 70, 14, 60, 61, 97, 30

Ans:

[40, 20, 70, 14] [60, 61, 97, 30]

[40, 20] [70, 14] [60, 61] [97, 30]

[40] [20] [70] [14] [60] [61] [97] [30]

[20] [40] [14, 70] [60, 61] [30, 97]

[14, 20, 40, 70] [30, 60, 61, 97]

[14, 20, 30, 40, 60, 61, 70, 97]

8) Quick sort

Input : 78, 23, 31, 88, 43, 55, 67, 55

$\boxed{23, 31, 43, 55, 67, 55}$ + $\boxed{78}$ + $\boxed{88}$

↓

$\boxed{23}$ + $\boxed{31, 43, 55, 67, 55}$

↓

$\boxed{23, 31}$ + $\boxed{43, 55, 67, 55}$

↓

$\boxed{23, 31, 43}$ + $\boxed{55, 67, 55}$

↓

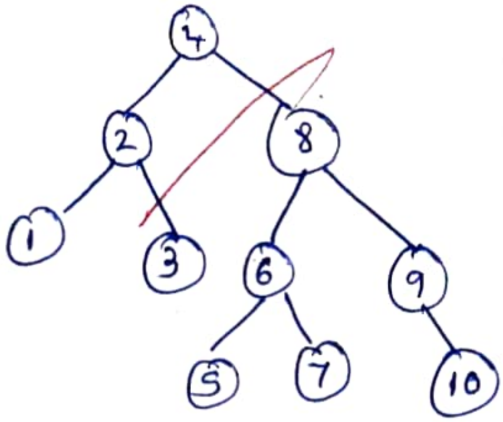
$\boxed{23, 31, 43, 55, 55, 67}$ + $\boxed{78}$ + $\boxed{88}$

↓

Sorted result : 23, 31, 43, 55, 55, 67, 78, 88

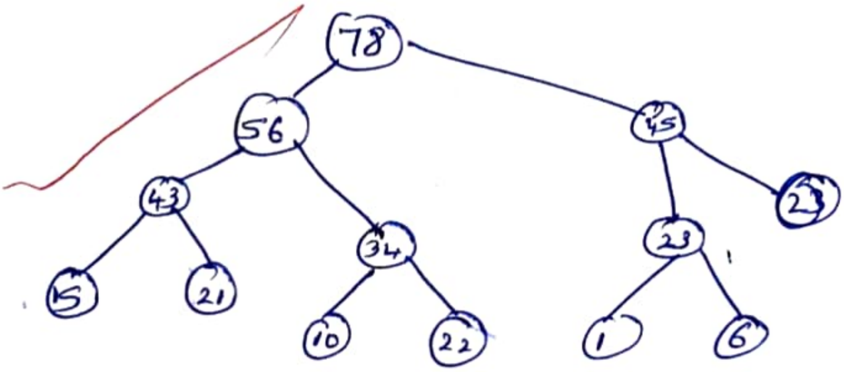
4) AVL TREE (insert 1 to 10)

Ans:



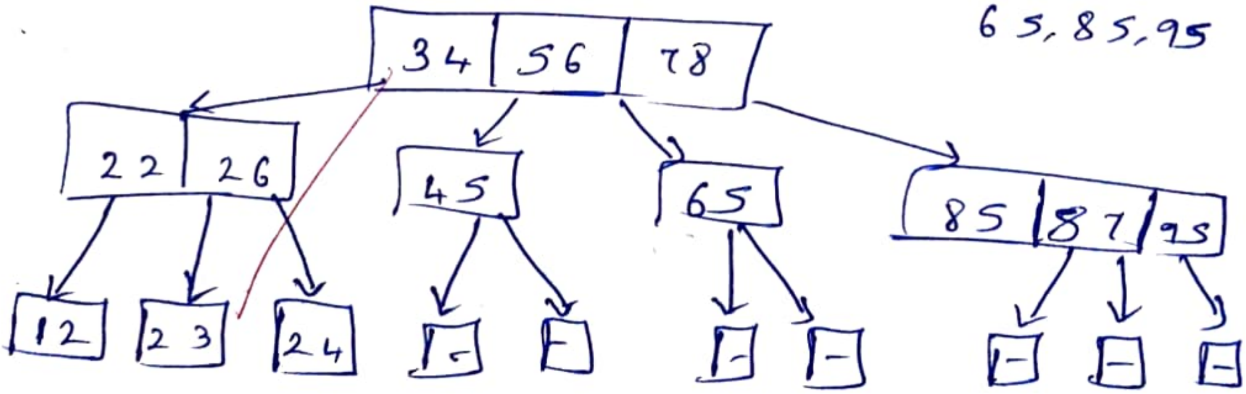
⑤ Binary Heap (Max Heap)

Ans: Insert: 10, 15, 12, 23, 21, 1, 6, 43, 34, 56, 78, 23, 45



⑥ B-TREE of ORDER 3

Ans: Insert : 34, 26, 45, 12, 87, 56, 78, 22, 23, 24, 65, 85, 95



(Each node has max 2 keys and max 3 children)

9) OPEN Hashing & Linear Probing

Ans: Given keys: 0, 1, 81, 4, 64, 25, 16, 36, 9, 49

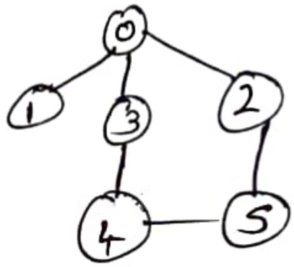
Hash function: $h(K) = K \bmod 10$

Hash Table (Size = 10)

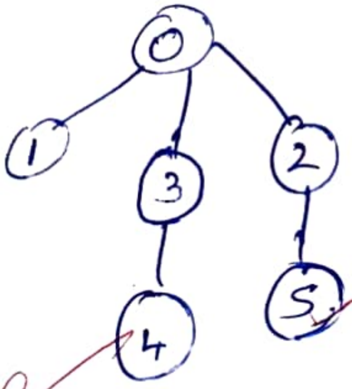
Index	0	1	2	3	4	5	6	7	8	9
Value	0	1	81	-	4	64	25	16	36	9*

→ 49 will be placed at next index 10 (if table size increased)

10) Consider the following graph & Construct BFS



Ans:



Step	Visited	Queue
1	0	1, 3, 2
2	1	3, 2
3	3	2, 4, 5
4	2	4, 5
5	4	5
6	5	-

BFS order: 0 → 1 → 3 → 2 → 4 → 5